



Top Intellectual Property Chain (White Paper V0.5)

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Forward	3
1.Introduction of Top Intellectual Property Chain	4
1.1 What is Top Intellectual Property Chain.....	4
1.2 Blockchain Big Data Algorithm Application and Data Flow Value.	4
1.3 Insist on the open source code	4
1.4 The core ideas	5
1.5 Prospects.....	7
2.Blockchain Technology Combine with AISA Apply to the Community.	7
2.1 TIP Community.....	7
2.1.1 The purpose of building community	7
2.1.2 TIP Community and TIP Points	8
2.1.2.1 The structure of TIP Community	8
2.1.2.2 Top Intellectual Point and Reward.....	9
2.2 The Digital Asset of TIPC.....	9
2.2.1 The creation of Encrypt Digital Assets of TIP.....	9
2.2.2 Advantage analysis of PoS V4.0	10
2.3 Blockchain with high performance and AISA access.	13
2.3.1 Top Intellectual Property Chain	14
2.3.2 High performance trading network of TIPC	23
2.3.3 TIP-AI	23
2.3.3.1 Artificial intelligence assistant.....	23
2.3.3.2 AI multi-platform connection.....	24
2.3.3.3 AI cross-domain diversion	24
2.3.4 BlockChain+ AI + IOT	25
3.TIP ecological structure	26
4. Community business model of TIP.....	28
4.1 CBusiness architecture	28
4.2 How do TIPC members get Top Intellectual Point	29
4.3 The selection process of high-quality intellectual property in the TIPC community	30
4.3.1 The vote.....	30
4.3.2 Transfer of IP ownership.....	31
4.4 Other application scenarios of Top Intellectual Point.	32
4.4.1 IP on the chain.....	32
4.4.2 Storage Permissions of TIPC	33
4.4.3 The Permission of Tracing and Viewing.....	33
4.4.4 Incubator	34
4. Management Standards of TIPC.....	34
Conclusion.....	35
Explanation of term.....	37
Reference	39

Forward

Since the 70's of last century, cryptography, distributed network, consensus plugin and computer hardware develop rapidly, and this provides a new plan for us to achieve institutional trust risk through multi agent environment, reduce transaction costs and enhance synergy effect through technology.

Block chain provides more innovative ways for cooperation, and more win-win opportunities to mutual trust node. The new mode of multi-party participation and peer-to-peer cooperation has gradually highlighted the value. The characteristics are multi-equal participation, intelligent collaboration, professional division of labor and value sharing, etc. Nowadays, Blockchain technology has shown certain potential in different fields.

At present, the most famous projects in the commercial application of block chain, such as Bitcoin and Ethernet, have fully realized the essence of technology sharing and management transparency in distributed business. Benefit from their pioneering role, the distributed technology based on open source code has been greatly developed.

The TIPC team has built a high performance TIP chain based on the existing block chain open source technology, thus, it has wonderful performance whether in the consensus mechanism, the accounting security strategy or the transaction performance, which provides more suitable development for the increasing demand of block chain transaction. The TIPC smart contract with safe and efficient can provide a fast and reliable trading environment for more and more complex applications. It constantly improve and surpass itself base on the demand of investors to achieve and maintain the high - performance block chain in the world.

We will actively develop the Blockchain technology in the worldwide, and create a very new concept of digital community with collecting Blockchain technology and ideas base on the high-speed and high-efficient TIPC. The real block chain community combines block chain technology with the innovation ideas, encouraging the whole people to enthusiasts the block chain technology and offers suggestions for it. Start with one node at a time, and gradually make the TIPC innovation community become the most influential block chain community in the world.

1.Introduction of Top Intellectual Property Chain

1.1 What is Top Intellectual Property Chain

Top Intellectual Property Chain, referred to as TIPC, is a new type of block chain +AI basis platform, based on block chain AI technology, which supports cross chain transactions and multichain interaction, providing trusted, safe, efficient and high performance block chain services. Community is the foundation of our existence. The purpose of the community is to have direct interaction between the builders and the members of the community, and the use of AI technology enables community members to participate fully in the construction of communities or projects. Both professionals and amateurs can brainstorm in this community, we will transform the understanding and ideal of block chain technology into a touchable idea.

Since collaboration between existing innovation teams is so inefficient, we can make it easier for entrepreneurial teams to find complementary teams through AI technology, which allows the choice diversified and easier to effective docking to realize the win-win cooperation. We build this community for people of common interest and members in here can receive dividends from community development and individual community contributions.

1.2 Blockchain Big Data Algorithm Application and Data Flow Value.

TIP focus on the future Big Data and the Blockchain's algorithm application for comprehensive development. With the continuous accumulation of "chain" data, it will present a certain kind of data aggregation and dispersion.

TIP is combined with AISA, which is combined with Micro-data and Big Data, and this whole is responsible for organizing all the "chain" and the corresponding "chain" data. The entire TIP community will grow and evolve as the AI algorithm development. At the same time that the data volume and data value are increasing, TIP is also expected to obtain the value growth of data flow addition, and increase the return rate of long-term investment and the willingness of users.

1.3 Insist on the open source code

Unlike the traditional trust mechanism which based on single credit endorsement entity, the trust mechanism of block chain is the common trust of multiple participants on transparent and trusted rules, and the trust of objective information technology. Therefore, in order to increase trust and full transparency, and reduce or even eliminate human intervention completely, the

Blockchain has appeared as an open source technology from very beginning, and most of the Blockchain technology platforms also exist as open source communities.

The essence of open source is sharing technology, and its meaning is that: Create a multi-stakeholder community through open source sharing technology and promotion technical standards. With the popularity of open source technology and the increasing number of participants, the division of labor in the ecology will be more detailed and the impact range will be wide. Thus, it is advantageous in competition with other similar technology of closed source. Build a Blockchain open source underlying platform, which is understandable, reasonable compliance, low access threshold, and easy to use. So that more interest can be driven and promotion effect can be more obvious. It also hopes to build an international technology ecosystem for sustainable development through the construction of open source community.

1.4 The core ideas



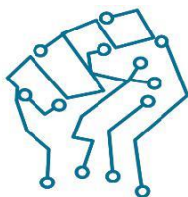
Distributed

Trusted public Blockchain distributed ledger, and high-performance distributed database cluster.



Real Name Based Public Chain

- Real-name registration.
- Guided by policies and regulations, with the development trend of Blockchain development.
- Real name mining machine, perfect risk control model.



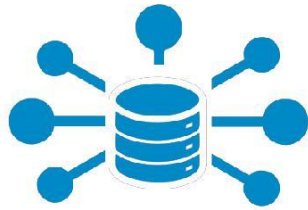
AI

The personal assistant with micro-data AI technology. User habit analysis, precise intelligent investment. It can be used to analyze users' habits, and it is an investment adviser with precise intelligence.



DAPP Platform

The professional and convenient DAPP development platform enriches the application types of the community and promotes the diversified development of Blockchain technology.



Big Data

The application of Blockchain big data algorithms is integrated and developed, with the AI technology evolves.



cooperation

Connect to the entrepreneurial team, and promote more win-win cooperation.



Professionalism

Professional technical teams support the operation of the community, and more senior Blockchain experts make the professional rating.



Cash Your Idea

To convert the excellent block chain IP directly into asset value, which can constantly attract and encourage people with innovative thinking and high-tech talent..



Freedom of speech

The members can communicate freely in the community, promoting the healthier development of the community so that a more secure and stable community environment can be created.



Eco-Development

The community has built a series of community ecosystems around high-performance TIPC and high-speed, secure database clusters, and build and expend the scale of the TIPC community.

1.5 Prospects

The TIP community is the first step in our joint development of Blockchain technology with Big Data and AI. Our driving force is the collision of brainstorm ideas of the members in the community, and it is our mission to collect the truth that is scattered among the masses. We will use micro-data AI to serve every member of the community, and the automatic evolution of Big Data will maximize the collective interests of the community. To distribute the contribution to the masses is our attitude and promote the Blockchain technology development is our dream!

2. Blockchain Technology Combine with AISA Apply to the Community.

2.1 TIP Community

2.1.1 The purpose of building community

In recent years, block chain technology is widely known as the soaring price of BTC, more and more ICO projects appear with the intelligent contract of the Ethereum. The appearance of DAPP technology is more convinced that Blockchain technology is must to be the steeping stone of the next generation of revolution. Because of this, many individuals or groups around the world who pursue technology or sensing economic profits have launched their own Blockchain related services and products. However, It is obvious that the Bockchain should be peer-to-peer, and it is also a natural chain service for the masses.

In today's society, truth is often buried in the masses. The purpose of building TIPC is to excavate potential ideas from the masses and set up the most direct platform for ordinary people to transform thoughts into values. TIPC will reasonably allocate available resources, scientific management communities, mining and guiding each community member to their potential value, by organize the micro-data of community users' habits , as well as the overall integration of the community ecology big data.

The IP of TIPC provides authoritative community assessment and professional guidance to convert quality idea into professional Intellectual Property. Through the analysis of Big Data, guiding the entrepreneurial team to develop together, and improving the cooperation efficiency between professional teams; Through the influence of the community platform to attract investment for outstanding IP, accelerate the value realization of outstanding innovation, and realize the win-win situation.

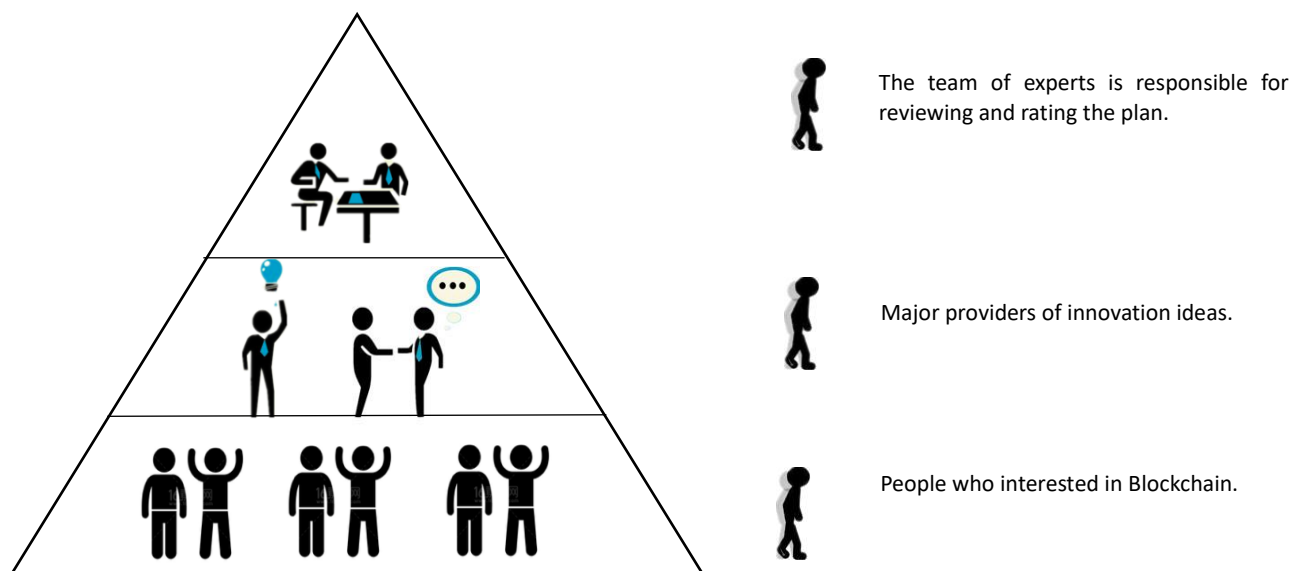
2.1.2 TIP Community and TIP Points

2.1.2.1 The structure of TIP Community

The community structure of Top Intellectual Property Chain is divided into three layers, which is similar to a pyramid shape. The bottom layer is the people who are interested in the Blockchain. The middle layer is the innovation idea provider for the Blockchain, and the Top layer is the expert team.

The bottom of the pyramids is the largest civilian review group who provide ideas, and it is also has a large network. They give the most appealed comments at the same time, while they can use the most simple language to describe the most desired for Blockchain application, bringing fresh inspiration to the whole community.

The middle layer of the pyramid is some people who have professional knowledge or block chain practitioners, or the scholars who study the Blockchain technology deeply. The Blockchain work that they do gives them a higher view than the ordinary people, so they have deep and unique opinions on the innovative application of block chain. Therefore, the members of this layer will become the main force of the idea proposer of innovation Blockchain in the community.



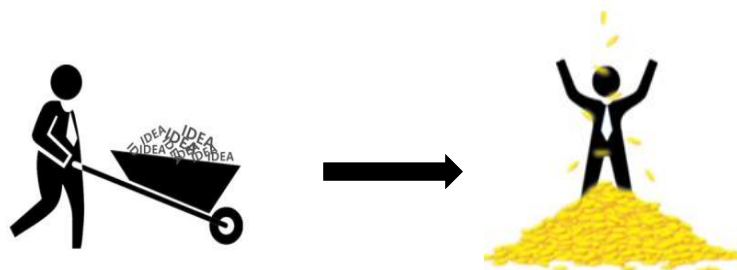
2.1.2.2 Top Intellectual Point and Reward

It is well known that the most popular Blockchain application is BTC and its ecological chain obviously, no matter investors or miners have deep interest for it. The reason is obvious, because BTC brings the most immediate benefits.

The community is aimed at the general public. Certainly that it has a competitive incentive system. This will not only attract more people to build the TIP pyramid, but also inspire the masses to contribute more valuable ideas to promote the development of the whole block chain technology.

TIPC, which owns a strict idea rating system and corresponding reward mechanism. It is based on the Blockchain technology, and has issued a set of virtual digital currency-- Top Intellectual Point (TIP), which is used to reward and invest in the chain.

Innovative idea contributors can obtain the corresponding TIP award through expert review and popular vote, while investors holding TIP can obtain the use rights of high-quality idea through the transfer of assets. In this way, the process of brain storming into property is not only the self-realization of idea contributor, but also a good opportunity for investors to find quality projects. Win-win is the foundation of the TIP community.



2.2 The Digital Asset of TIPC

2.2.1 The creation of Encrypt Digital Assets of TIP

TIPC adopts hybrid packaging mode of PoW+PoS:

TIPC digs Top Intellectual Point by early POW(Proof Of Work). This is the stage of construction of TIPC, which only allows trusted node with cooperative credentials to complete. The TIP asset is generally collected centrally by TIPC's special account to ensure the justice and fairness of the community. Meanwhile, each block is searchable to ensure the asset

transparency of the TIPC community.

PoS(Proof of Stake) is used in the formal operation of the community. At this stage, PoS is a enhanced version of PoSv4.0, which is innovated on the basis of the most advanced version 3.0. It strengthens the performance of the anti-attack and enhances the security of the asset. It guarantees to initiative of the miners while strengthens the anti - attack performance of the chain , and improves the security of assets.

2.2.2 Advantage analysis of PoS V4.0

The fundamental of design of PoS node packaging competition mechanism is, on the premise that blockchain run smoothly, ensuring the safety of assets on the chain, ensuring sufficient incentive mechanism for miners, and following the principle of environmental friendliness, which lays the foundation in order to build a sustainable ecosystem for TIPC.

Problems with PoS V1.0:

- 1) The coin age is malicious abused by the node, which can obtain very high network weight, thus launching double-spending attack.
- 2) The system design that encouraging nodes do not need to keep online for long period to get benefit, but can accumulate coin age when it shut down. And turn off the device immediately after it is get profit when it turned on.
- 3) The components of PoS algorithms are predictable, which This brings security risks to the system.
- 4) An output transaction will be divide into two parts through create a new deal when burn down coin age, it results fragmentation for most outputs and large transfers may exceeding the upper limit of the block space, which lead the transaction cannot be completed.
- 5) Bookkeeping methods will become more and more centralizing, and even a node may has more than 51% of the equity in the future.
- 6) The rights and interests will be increased randomly as the transaction is packaging, which is not consistent with the general equity incentive mechanism in reality.
- 7) Calculating power attack. This means that the rights and interests of the possessor do not match the actual income. A node can gain benefits that is far higher than its interests by investing the calculating power that above average.

The formula for calculating the probability of packing permission of node block in PoS V1.0:

$$\text{proofhash} < \text{coins} \cdot \text{age} \cdot \text{target}(t) \quad (1)$$

proofhash-- It is the hash value that gets the block package right, depending on the stake modifier and the unused output and the current time.

coins--It is the amount of available coin in the current node account that participate in the competition of packaging right.

Age--The amount of available coin that is mentioned above dormant.

Target(t)-- It is the packaging difficulty of the whole blockchain, which is function of time.

Eq. (1) is the initial model to calculate proofhash by PoS. It Depends on the amount of coin held in the account and its dormant period determine the probability that the node of the current account gets the block package right. Proved by practice that its bad for the miners' nodes online, and there is a high risk of illegal attacks caused by the coin age accumulation.

$$\text{proofhash} < \text{coins} \cdot \text{target}(t) \quad (2)$$

Eq. (2) is the calculation formula for calculating proofhash after eliminate the influence of coin age on the packaging power of the node since PoS V2.0/V3.0. This could encourage more miners to stay online, which bringing more stability and security to the entire blockchain system.

The packaging competitiveness of PoS V4.0 created by TIPC originates from the weight of the account holding amount in the total amount of coins issued by the blockchain system. It is different from traditional PoS consensus algorithm in a competition way of coin age plus coin weight, the innovative consensus algorithm of PoS V4.0 of TIPC is not only keeps the miner's incentive mechanism perfectly, but also totally avoid the risk of traditional PoS consensus algorithm with 51% attack, which provides a safe operating environment for the community.

PoS V4.0 continues the PoS V3.0 that abandon coin age calculation and the block of calculate risk in advance, meanwhile, inherited block rewards system of PoSV3.0 ---provides a steady consensus foundation while restrains the non-active nodes.

PoS V4.0 is especially applicable to the registered and issued mining cluster or real-named node. The formula (2) is improved by the original modulation factor of SMPM (Stake Minting Probability Modifier). Under the modulation factor constraints, proofhash restrains coin holding weight of node account and the package probability of corresponding block. Thus, it can avoid the risk of 51% attack on the whole blockchain system due to its excessive weight of holding coins.

Calculation formula of PoS V4.0 consensus algorithm:

Take the BlackCoin public chain V2.12.1.0 as an example, Introduce modulation factor SMPM(SMPM= $f(w_x)$):

$$f(w_x) = \begin{cases} kw_x & (0 \leq w_x \leq Default) \\ e^{\frac{(w_x - Default)^2}{2}} - e^{\frac{(50\% - Default)^2}{2}} & (Default < w_x \leq 50\%) \end{cases}$$

In above equation, w_x is the percentage of the total amount of coins in the blockchain system that is held by the account holder, also known as the currency weight. x is the number of accounts in the block chain system so that calculation formula of w_x is: $w_x = \frac{coins_x}{\sum_{x=1}^{x=all} coins_x} * 100\%$. And $f(w_x)$ is the mathematical expression of SMPM.

The default is the best proportion point of SMPM method and k is the slope of the line when $w_x \leq Default$, both of them can be adjusted according to the assets overall of the blockchain system.

$$\begin{cases} P_x(w_x, t) = \begin{cases} \alpha(t) \cdot w_x & (0 < w_x < Default) \\ \beta(w_x, t) \cdot f(w_x) \cdot w_x & (Default < w_x < 50\%) \end{cases} \\ \int_{t=0}^{t=1day} \sum_{x=1}^{x=all} P_x(w_x, t) dt = 24 \cdot 60 \cdot 1 \\ proofhash < w_x \cdot f(w_x) \cdot target(t) \end{cases}$$

The proofhash is the hash value of the next block to be generated, α is function of time t and β is function of w_x and time t .

At some point of time, the account's w_x is between 0 and Default, when the probability of getting the rights of package the next block of that account's P_x is proportional to w_x . That is to say, our patent method encourages all node to hold as much money as possible when account of coins that is less than the Default, by adjusting the probability of obtaining the package rights for the next block. Further more, when w_x is between $Default \sim 50\%$, the probability of getting the rights of package the next block of that account's P_x related to w_x and is also restricted by SMPM. As w_x increases gradually in the $Default \sim 50\%$ range, its corresponding SMPM will gradually approach 0, thus achieving the probability of adjusting w_x to get the rights of package the next block in excess of Default point.

All in all, our patented method does not encourage any single account to hold lots of coins. Assuming that the account's w_x exceeds 50%, that the probability of getting the rights of package the next block strictly equal to zero, so that, our method can completely avoid the 51% attack risk caused by excessive coins accounts.

The calculation expression of SMPM in this patent method is that:

$$\left\{ \begin{array}{l} f(w_x) = \begin{cases} kw_x & (0 \leq w_x \leq \text{Default}) \\ e^{-\frac{(w_x - \text{Default})^2}{2}} - e^{-\frac{(50\% - \text{Default})^2}{2}} & (\text{Default} < w_x \leq 50\%) \end{cases} \\ \lim_{w_x \rightarrow 50\%^-} f(w_x) \rightarrow 0^+ \\ \lim_{w_x \rightarrow 0^+} f(w_x) \rightarrow 0^- \end{array} \right.$$

The optimum value of *Default* is 10% and $k=1$, both of them can be adjusted according to the assets overall of the blockchain system.

According to the expression of SMPM that when the weight of account (w_x) is reaching to the left of 51% in any node of any account, the SMPM modulator $f(w_x)$ is infinitely approaching right to zero(0). in other words, the qualification of packaging block will be lost if the amount of coins that account holds more than 51%. And that $f(w_x)$ will be infinitely approaching right to zero(0) as the weight of account $w_x()$ is reaching to the right of zero(0), which all nodes in SMPM modulation factor.

Summarize the above, Calculation formula of PoS V4.0 consensus algorithm:

$$\left\{ \begin{array}{l} \text{proofhash} < w_x \cdot f(w_x) \cdot \text{target}(t) \\ f(w_x) = \begin{cases} kw_x & (0 \leq w_x \leq \text{Default}) \\ e^{-\frac{(w_x - \text{Default})^2}{2}} - e^{-\frac{(50\% - \text{Default})^2}{2}} & (\text{Default} < w_x \leq 50\%) \end{cases} \\ \lim_{w_x \rightarrow 50\%^-} f(w_x) \rightarrow 0^+ \\ \lim_{w_x \rightarrow 0^+} f(w_x) \rightarrow 0^- \end{array} \right.$$

2.3 Blockchain with high performance and AISA access.

Although the popularity of Blockchain technology continues to rise, the biggest problem we directly face is low handling capacity of transaction. The Blockchain system with open source code of the high concurrent transaction ability is generally low. The bitcoin community is trying to increase the transaction processing ability to a certain extent by block expansion, SegWit technology, etc, but these methods can not lead to the transaction processing ability of magnitude improvement.

Due to the largest block chain open source project-- BTC and Ethernet have only one main chain, and all functions and data are added to the main chain, resulting in rapid expansion of blocks. The large block volume and the long synchronization time lead to the current open source

Blockchain technology , especially in the aspect of enterprise application, its transaction concurrency capability, data storage capacity, versatility, functional completeness and ease of use are all insufficient obviously. Based on the consideration of transaction performance, capacity scale, privacy protection and compliance supervision, thus, the volume processing capacity of block chain technology still could be improved greatly.

With the appearance of the block chain creation block, the number of the whole Blockchain system will get bigger and bigger, and eventually the data flow will be formed by Continuous growth and data showing a high correlation, traceable, unfalsifiable. If the distributed storage is still only found in the distributed books, the possibility of practical application of this technology will not enough to let public has optimistic view. That is why it is necessary that the comb data cleaning, data flow analysis, and user guide auxiliary micro data. It also will become a necessary trend for such a huge data volume, user system

2.3.1 Top Intellectual Property Chain

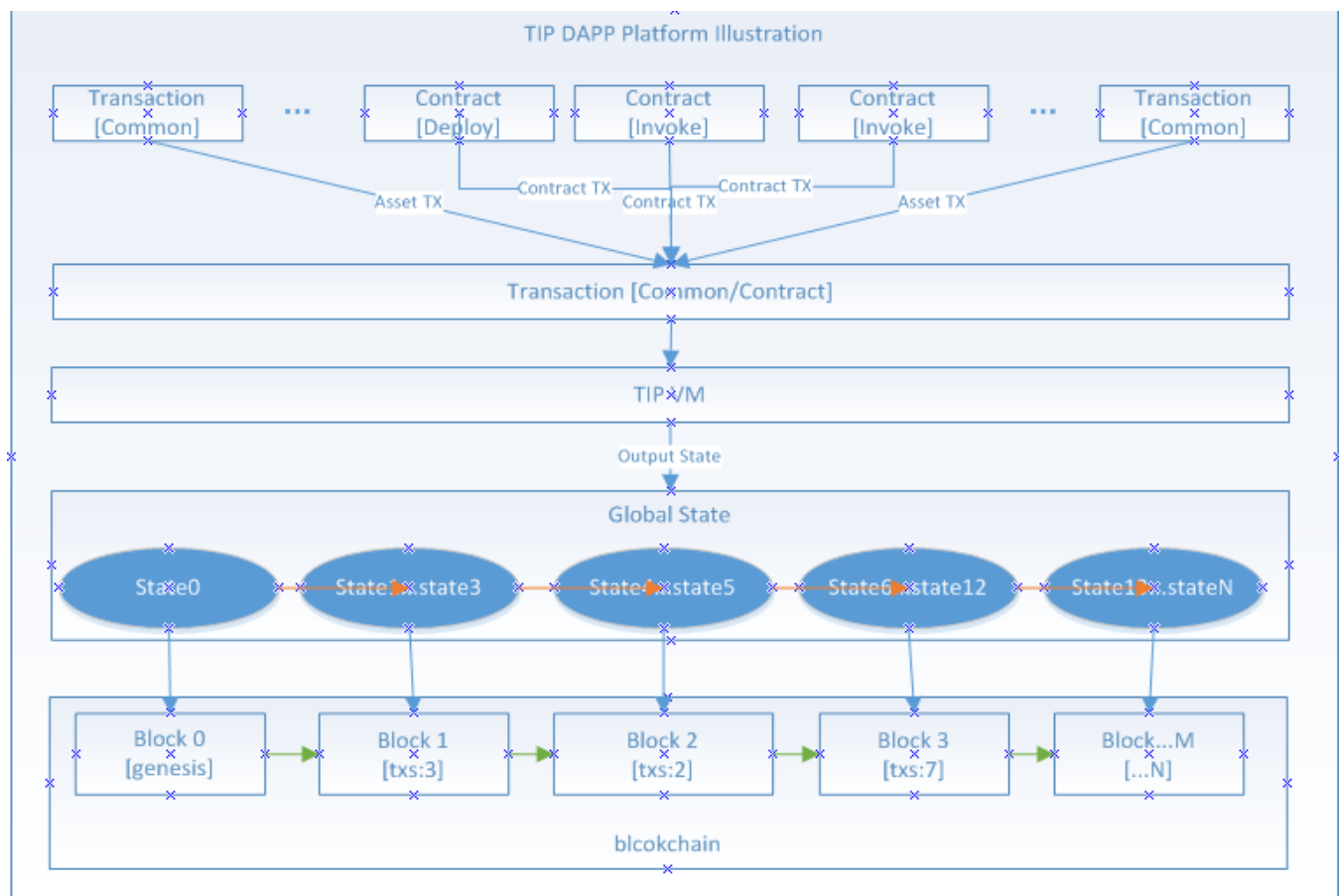
TIPC is a kind of dual-core data analysis engine that integrates Big Data and micro-data to provide reliable, safe and efficient high performance block chain service for cross-chain transaction and multi-chain interaction.

The main functions are listed as follows:

(1) DAPP platform of TIP



DAPP platform of TIPC is a one-stop development platform for Blockchain, it encapsulates the underlying Blockchain technology, combining with the Micro-data AI and Big Data development interface, which provides comprehensive Blockchain technology development services. All members of the community can develop their ideal block chain application products through the convenient way. It also can reduce the complexity of the project start from scratch to develop a complete Blockchain. This could let the Blockchain application developer develop directly based on the platform. Developers just need to focus on the development of the application itself and reduce the difficulty of application development due to the complex logic of the Blockchain and AI technology. DAPP platform has the advantage of providing Blockchain specific functions of data confirmation and value transfer, which reduces the obstacles of application of AI, Big Data and micro-data technology. DAPP platform has great advantage in transaction security, change of industry production relations, reduce the cost of operation and maintenance, and reduce the cost of technology development, etc.. The application of Microdata AI and Big Data can also greatly improve the user experience.



DAPP platform is based on the high performance global commercial grade TIP VM, and the high-speed and efficient assets and smart contract transactions are all connected here. The use of

asset transactions and smart contracts is implemented on chain, and it can be traced back to provide safe and reliable guarantee. The development platform of TIP DAPP providing excellent support for AI/IOT technology, it is really suitable for a new generation of Blockchain system integrating AI and IOT.

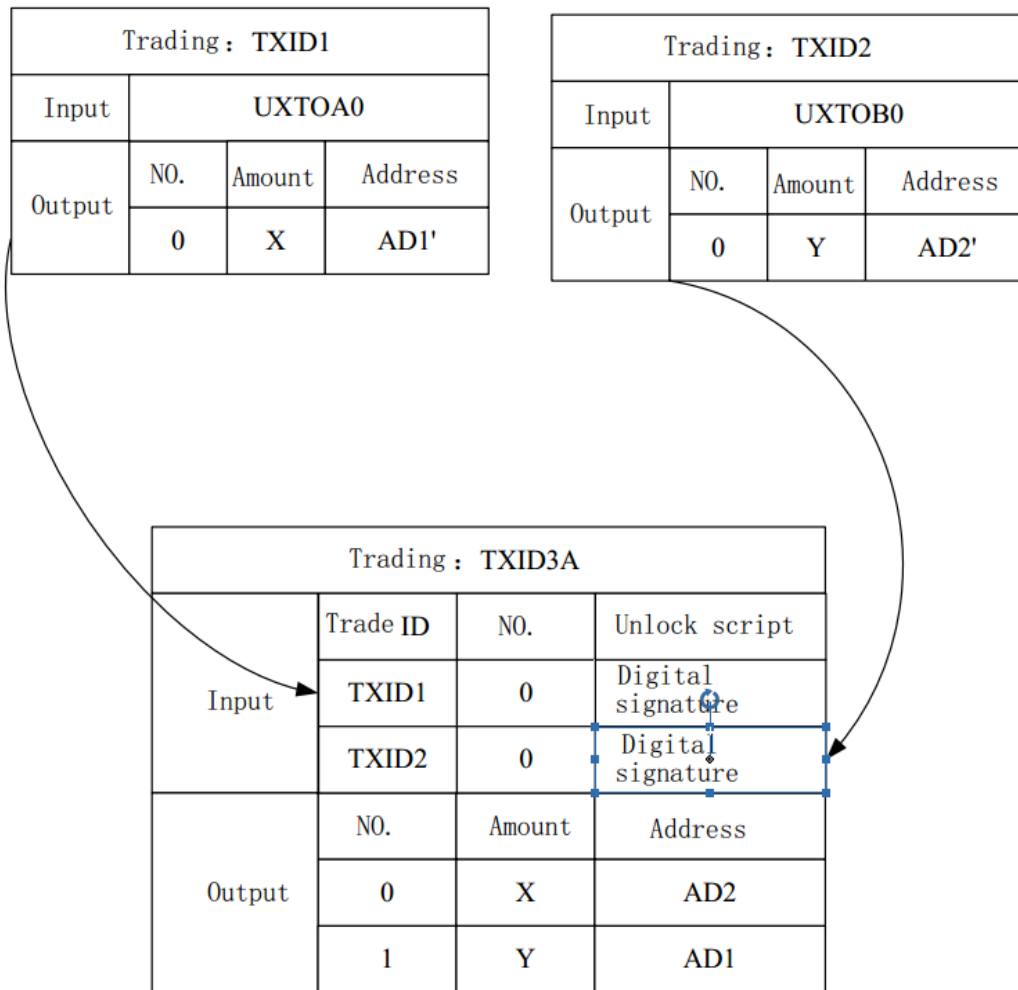
(2) Cross-chain Transaction

There is no third party participation mechanism between different kinds of digital assets, and it guarantees users' assets safety without any dependence on centralized platform, while provides the best service availability. Cross chain transaction of TIPC supports two-way transactions between logical chains, and smart contracts with multiple signatures initiated by logical chains to achieve one-on-one cross-chain transactions without third party participation.

TIPC cross-chain trading patent scheme is: By forming multiple logical chains on a physical Blockchain, and each logical chain has a corresponding identity in the transaction list, making different logical chains record different types of digital assets, and it can be traded freely within the logical chain. At the same time, building a cross-chain transaction when make a cross-chain transaction.

The input of the cross-linked transaction sheet consists of two or more identifications of different types of digital assets (different types of unexpended outputs), The output consists of receiving the address of the corresponding asset on a different logical chain. Thus, cross - chain transaction lists build a cross - chain joint transfer. Therefore, the cross-linked transaction sheet is verified and recorded on the physical Blockchain to realize the conversion and transfer of digital assets across the chain.

(This cross-chain trading patent method shall not be used in any form without authorization)



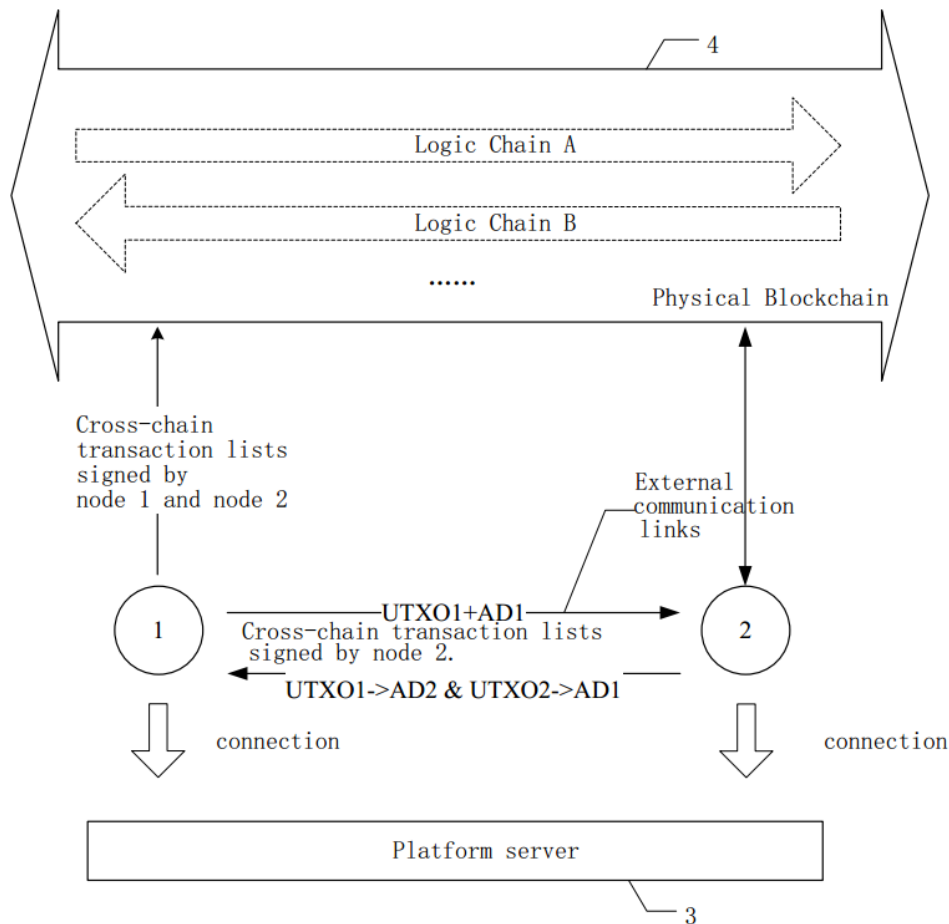
The input and output diagram of Cross-chain Transaction

Cross - chain transaction with two-node. The first node is expected to transfer the identity of the first unspent transaction output (UTXO) to the second node, and send it to the second node for receiving the first address of the price digital asset. Past that, the first UTXO is the first type of digital asset recorded in the first logical chain; the first address is the address of the first node in the second logical chain; the transaction sheets of the first logical chain and the second logical chain are recorded in different forms on the physical Blockchain respectively.

Accordingly, the second node constructs a cross-chain transaction sheet. In this, the input of the cross-chain transaction list includes the first UTXO and the second UTXO; the output of the cross-chain transaction sheet includes the second address and the first address. According to this, the second UTXO is the second type of digital asset recorded in the second logical chain; the second address is the address of the second node in the first logical chain.

The second node sends the cross-chain transaction single digital signature to the first node;

The first node digitally signs the cross-chain transaction sheet, and it will includes the digital signature of the first node and the second node that broadcast to the physical Blockchain network.



The diagram of patent method for Cross-chain Trading.

(3) Multiassets Transaction and IP ecosystem

TIPC allows many kinds of assets exist at the same time in a same public Blockchains, and transaction and settlement are independent of each other. It is easier to diversify the assets and manage the security of funds. Thus the TIPC can provide better Blockchain interaction schemes for the TIP community.

TIP are supported as first generation in the TIPC, which allows each IP projects to issue their own digital assets on TIPC B.

Blockchain via smart contracts, new digital asset's name and its exchange rate with TIP can be submitted here also. And the one's logo can be submitted by wallet of TIP.

Startup company may build their own subchain by apply to the TIP community and consume some TIP points. Each subchain could introduce interfaces of smart contracts, Micro-data AI & Big Data and storage space of high-performance database clusters, which makes the IP projects to form a perfect ecological of artificial intelligence and Blockchain technology.

Multi-asset operation and management are independent of each other, which facilitate the circulation , sorting of assets and mutual interaction among assets. Therefore, the value of services provided by TIPC is much higher than that of the single token mode issued by Ethereum.

The subchains of IP projects depend on the ecology of the entire TIP community With the help of Micro-data AI & Big Data algorithms, which is conducive to artificial intelligence to summarize micro-data of members of each IP projects and the big data of the whole TIP community, so that, a more explicit state of supply and demand is formed at the TIP community level.

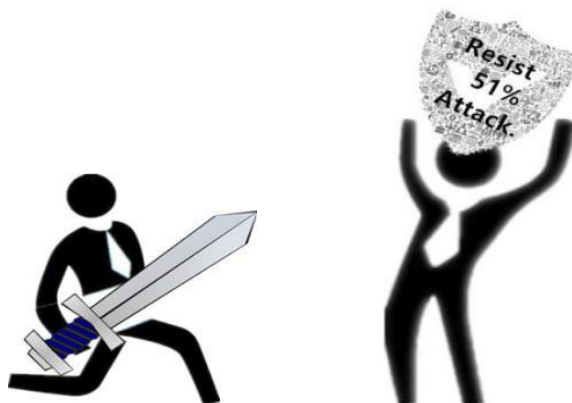
The TIP community can provide complementary cooperation among Startup companies, which improves the efficiency of supply and demand between them and promotes a healthier and more coordinated development for the whole community.

(4)Prevent 51% attacks at the PoS stage(PoS v4.0)

TIPC adopts PoS (Proof of stake) V4.0 which is more energy conservation and environment protection, to competitive block packaging right and asset appreciation program. In the traditional Pos mining algorithm, it will has safety loophole to the history book or the funds security if a single account of a single account is more than 51% of the current Blockchain system. It is the first time that Top Intellectual Property Chain optimize the packaging Competition at PoS stage by patent algorithm:

In the public Blockchains with real name , under a certain condition, the PoS packaging probability will be adjusted by adjusting the amount of coin holding in the account in any node, and encouraging the miners to actively participate in the operation of the node may completely avoid the issue of 51% of the attack caused by the excessive weight of the currency. This creates an unprecedented security environment for TIP community assets.

(Details of the specific patent algorithms will be published after the patent application cycle ends, any plagiarism in this period will be held accountable.)



(5) Higher performance for transaction processing

The consensus algorithm is an important aspect of conditionality performance, and the typical consensus algorithm in Blockchain project are: PoW、PoS、DPoS、PBFT,etc. Their performance are:

Systems		Committee Formation (Resource)	Performances	
			Throughput	Latency
Hybrid	ByzCoin	Pow	1000 tx/s	10-20s
	Algorand	Lottery	90 tx/s	40s
	Hyperledger	Permissioned	110 tx/s	<1 s
	RSCoin	Permissioned	2k tx/s	<1 s
	Elastico	Pow	16 blocks in 110s	110s for 16 blocks
	Omniledger	PoW/PoX	10k tx/s	1s
	Chainspace	Flexible	350 tx/s	<1 s
proof-of-X	Ouroboros	Lottery	257.6 tx/s	20s
	Snow-white	Stake	100-150 tx/s	-
	Intel PoET	TH12	1000 tx/s	-
proof-of-work	Bitcoin	Pow	7 tx/s	600s
	Bitcoin-NG	Pow	7 tx/s	<1 s
	DECOR+HOP	Pow	30 tx/s	60s

Another factor that restricts the performance of Blockchain transaction is the structure of the ledger. At present, the typical ledger is designed as a single chain structure, and all transactions can only be packaged and processed in order, and the transaction lacks the ability of parallel processing.

PoS v4.0 that TIPC using can make the transaction throughput reach to 100 thousand per second, which can be on a par with the middleware buffer queue processing capability of traditional centralized platform. This has laid a solid foundation for the continuous development and expansion of the TIP community and the participants. Meanwhile, this is also a key prerequisite for the development of the TIP community. The principle is that, base on the smart contract, the subsidiary ledger which relies on the local node consensus algorithm and the general ledger which relies on the global node consensus algorithm is freely switched. In the meantime, the transaction results are recorded on the Blockchain to ensure the untamperability and verifiability of information storage or asset liquidation. TIPC can use smart contract establish asset transfers network among multiple nodes, therefore, it can meets the transfer requirements of asset with accurate, fast, efficient and reliable, and supports high frequency and fast trading with zero loss of assets, such as energy.

(6) Distributed Storage

TIP uses the hash algorithm to list the excellent Intellectual Property that has been selected as a professional document and to calculate the hash value in this entire file with the hash algorithm. Store and multiple copies of documents in a professional high-speed distributed

database; Store the hash value in the block of Top Intellectual Property Chain; Meanwhile, set up a high - speed database index to establish a strict correspondence between the document hash value and the Blockchain, in order to be truly traceable and searchable. The TIPC community of TIP can truly respect every idea, rigorously treat every KB data, and strive to become a completely trusted Blockchain community.

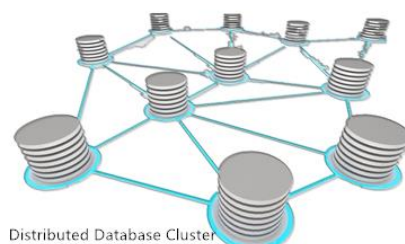
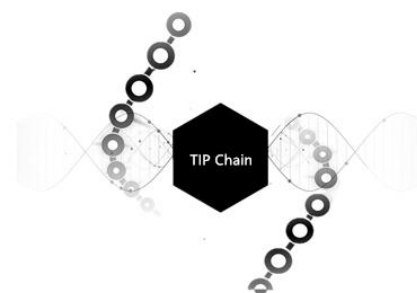
TIPC establishes a professional, high-speed and advanced distributed database cluster, we call this cluster as HPDC (High Performance Data Cluster). HPDC let the account address in Merkle DB of block database of TIPC map to HPDC cluster. Merkle DB is the database for the Blockchain storage node account address and block information. The high quality IP hash that mentioned above exists in this block information. The HPDC of TIPC takes this as an opportunity to combine the account address and IP with the hash value.

Its significance is:

(1) It solves the defect of small capacity of block chain block. Mass storage and multiple backup of HPDC, which is not only unlimited expansion of the traditional block chain block size, to some extent, but also combining the data security of Blockchain with data backup of database cluster, it brings unprecedented data integrity and security guarantee.

2) Transaction information (IP hash value) in the blockchain is mapped to the HPDC data base, which allows the high speed and precision finding of the corresponding IP document according to the hash value, and solves the traditional blockchain query step complicated problem.

3) HPDC provides global advanced fulltext search capability, it is able to quickly match the IP in mass files according to any feature, so as to accurately correspond to the account in the block chain, which perfectly conforms to the needs of the community's business.



High-performance database clusters map to each other

(7)Smart Contract

TIPC do the optimum proportion of accept or reject in removing cross-chain function, for example, choosing a hash-locking to support the contract, or, reducing functionality to support maximum concurrency and high speed in the bitcoin open source project. TIP provides an advanced smart contract in Turing, while ensuring the throughput of trading volume, bringing technology to community members and make the user experience more fluent.

TIP VM(Virtual Machine) all over the world self-developed by our team, who ensures that the transaction is accurate and free of loss, not only improves trading efficiency, but also supports AI/IOT technology, which is a solid foundation for the next generation of blockchain technology compatible with artificial intelligence and Internet of things.



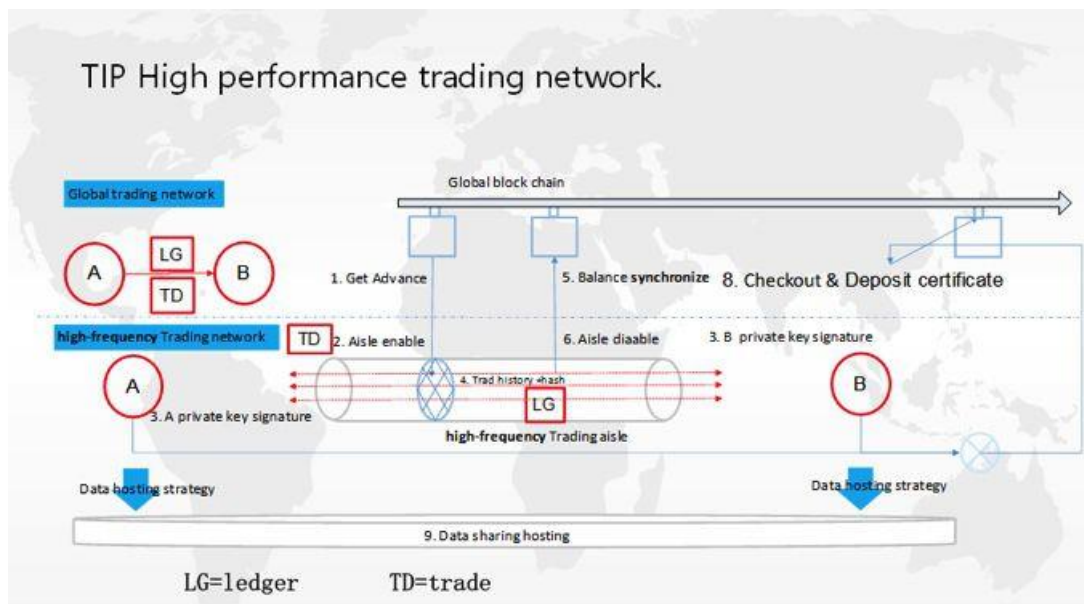
(8) Intelligent digital asset investment advisor

TIP innovatively use the AI technology on user's microdata for cross-chain diversion data, so that TIP could actively match the proportion of the appropriate digital assets, according to the user's personality habits and the micro data between different physical chain users in the community. Only with big data systems growth and self-learning day by day, can TIP community plan the portfolio of all member more accurately, which makes TIP to be a more clever and convenient for members. We aspire to create the world's top smart digital asset management blockchain platform.



2.3.2 High performance trading network of TIPC

1. The HFT is working on the high performance network of TIPC and it will be consistent with the global trading network through the "quota" and "balance", also , HFT is registered and trusted by means of "verification certificate" in the global network.
2. TD is essentially a smart contract to complete HFT."Open" and "close" high frequency trading channels with TD unit.
3. In the safe and closed HFT channel, HFT is triggered by the TX protocol, and each transaction is confirmed by both parties through the signature of A or B.
4. The TX information of A and B is stored on the global chain by the promissory verification rule.
5. A and B can make data sharing hosting based on their own data hosting policies.



The structure of TIPC

2.3.3 TIP-AI

2.3.3.1 Artificial intelligence assistant

TIP combines the block chain with AI innovatively, which allowing the artificial intelligence system to understand the members of the community better and makes the TIP-AI a top members' exclusive assistants. Every TIP-AI assistant is born to truly learn to get along with members of the community, actively caring for members and growing with them. IP-AI Assistant is not merely imitating or becoming a substitute for the user, but has its own independent emotional system. The use of original discrete dual master-nucleus calculus technology and Microdata micro-data

algorithm can give intelligent assistants a true personality and thinking mode. The assistants are linked with all TIP IP projects, giving community members the most accurate and useful recommendations and promoting the rapid and effective cooperation of startup companies in the community.

2.3.3.2 AI multi-platform connection

TIP-AI can exist in a variety of platform devices - mobile phones, computers, tablets, smart wearable devices (smart watches, accessories), smart speakers and so on. This gives the broader application development space for blockchain innovation projects, allowing each member to create innovative IP projects without the constraints of hardware and platforms. Community members can use TIP-AI to conduct digital point transactions and use in the most convenient and rapid manner for various devices. We will strive to link more devices and service platforms and eventually become the next-generation blockchain AI interactive ecological platform. The digital integral investment uses the pipeline to create a complete digital integration economic system.

2.3.3.3 AI cross-domain diversion

TIP-AI has cross-domain diversion capabilities and operates perfectly across TIP's cross-chain trading scenarios. All digital assets in TIP can use AI for personalized diversion. TIP-AI can understand the preferences and needs of each TIP community member, and through the development of big data, the more accurate and recommended investment trading portfolio.

List of TIP-AI key technologies:

①**Multi-Brain of the core field:** TIP-AI applies personal micro-data algorithms to deploy the core brain of the IP project in each IP project sub-chain. After the main core brain calculus enters the single-domain sub-brain, it continues to calculate the individual preferences. However, in different areas, the inter-brain TIP-AI can also achieve cross-domain connections, learn the relevance between different areas and gradually grow.

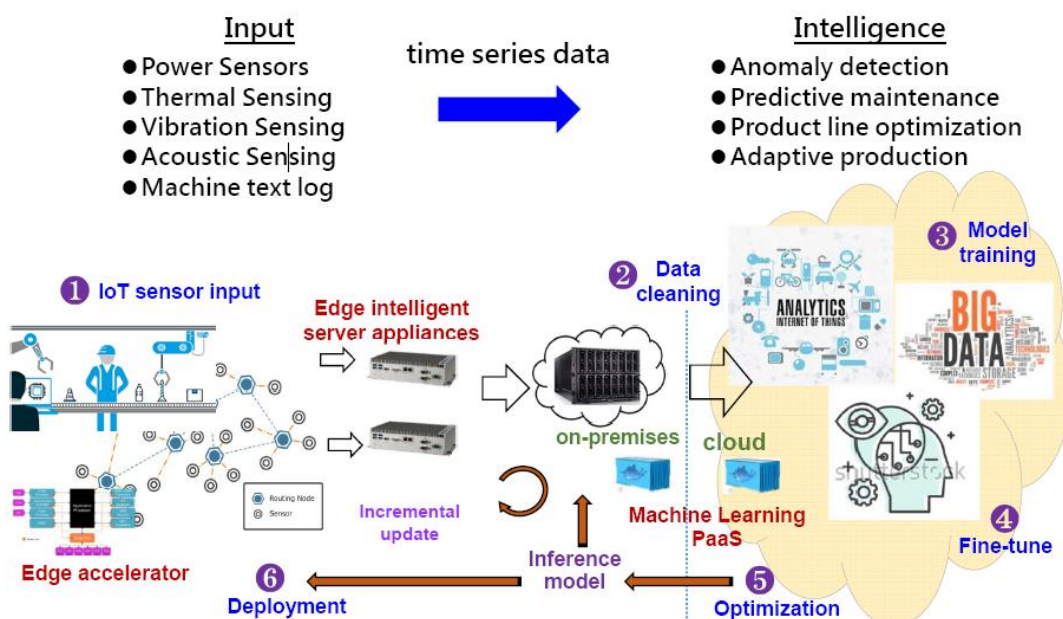
②**The Comma Field:** TIP-AI has the mechanism of autonomously generating a floating node in the middle node. It can constantly compare the associability of different IP sub-chains via data, create a floating slot, and then use the evolution mechanism. use and disuse Theory, so that the core of the TIP-AI master can have the concept of neural networks, the expansion of the slot to tens of millions to hundreds of millions of possible, more close to human thinking mode.

③**Sonar regression:** TIP-AI's unique "adaptability" initiative questioning and calculus mechanism can be extended to original sono AI. When TIP has new IP generation, TIP-AI can address cross-

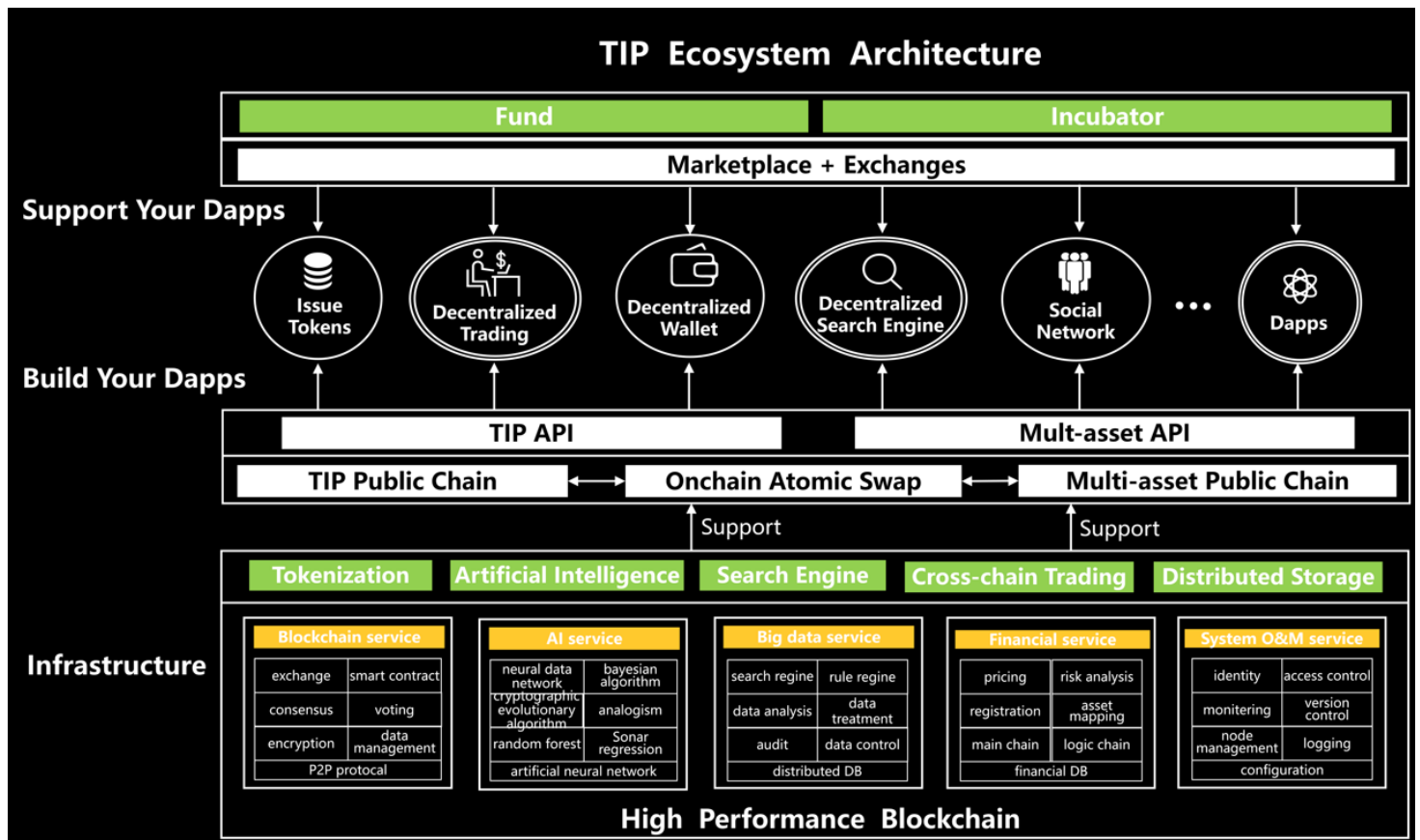
domain connectivity issues. Actively release the detection sonar, receive the user's response and correct the parameters, continuing the evolution of fitness. The difference between sonar AI and general AI lies in that it can actively create response parameters for self-correction requirements without having to passively find the value of parameters from existing data. As TIP community members continue to grow, TIP-AI will increasingly have higher value, and it is expected to solve problems that many existing AI algorithms cannot solve.

2.3.4 BlockChain+ AI + IOT

TIP Community Planning combines blockchain technology and AI technology to further introduce the concept of IOT into a comprehensive industrial chain such as smart wearables, smart cities, and smart homes. Really let the concept of blockchain connection be applied to create an all-round blockchain smart life. Extend the community of superior and intellectual chain community ecological circles, open the door for high-quality IP projects for community members, break down more hardware and software facilities, integrate technology and reality, and truly release community members' ideas to stimulate more. Good quality IP projects landed, better let blockchain, artificial intelligence technology benefit us.



3. TIP ecological structure



The TIP community ecological framework is divided into three layers as shown in the figure above: infrastructure layer, community application layer and fund incubation layer;

Infrastructure layer

The infrastructure layer is provided by the blockchain technology solution with TIP high-performance trusted blockchain public chain and distributed database cluster as the core. The basics of financial services and system operation and maintenance services are also deployed at this level. The high-performance trusted blockchain can provide basic functions such as basic warrants, traceability, and rights trading for IP services; high-performance database clusters can provide reliable storage, fast positioning, accurate search, and convenient management for massive IP.

Financial services provide professional and efficient asset certification channels for IP on the chain, and risk control of the entire community assets. The professional and convenient artificial intelligence service provides exclusive Microdata AI and Big Data services, which makes the entire TIP community intelligent. The operation and maintenance services ensure the community membership and access control while providing a stable operating environment for the entire TIP

community software system.

Community application layer

The community application layer provides developmental ground for the development and expansion of the TIP community. The activities of all community members will be directly reflected in this layer. The application interfaces of the basic layer blockchain technology and database clusters are included in the application layer, and community members can facilitate Quickly call interfaces for various functions to create their own unique blockchain application-level services.

The TIP-DAPP platform is the best embodiment of the application program. Community members can gradually build a dedicated IP DAPP according to their own IP needs, making IP faster and more convenient to achieve asset value conversion.

TIP digital wallet, members can use this wallet for professional asset management, both TIP points and IP credits issued by the members themselves are managed independently in this account, to ensure that assets are clear and precise combing.

The distributed search is based on the database cluster's search portal. The TIP community uses the most professional database technology to provide the most convenient services for the public, and massive IP fuzzy keyword queries and rapid query feedback.

Distributed transactions and high-performance distributed trading networks provide the community with up to 100,000 tps of non-destructive transaction throughput, ensuring the safe and rapid conduct of transactions and providing basic guarantees for the development of community business.

Other basic services are not listed here.

Fund incubation layer

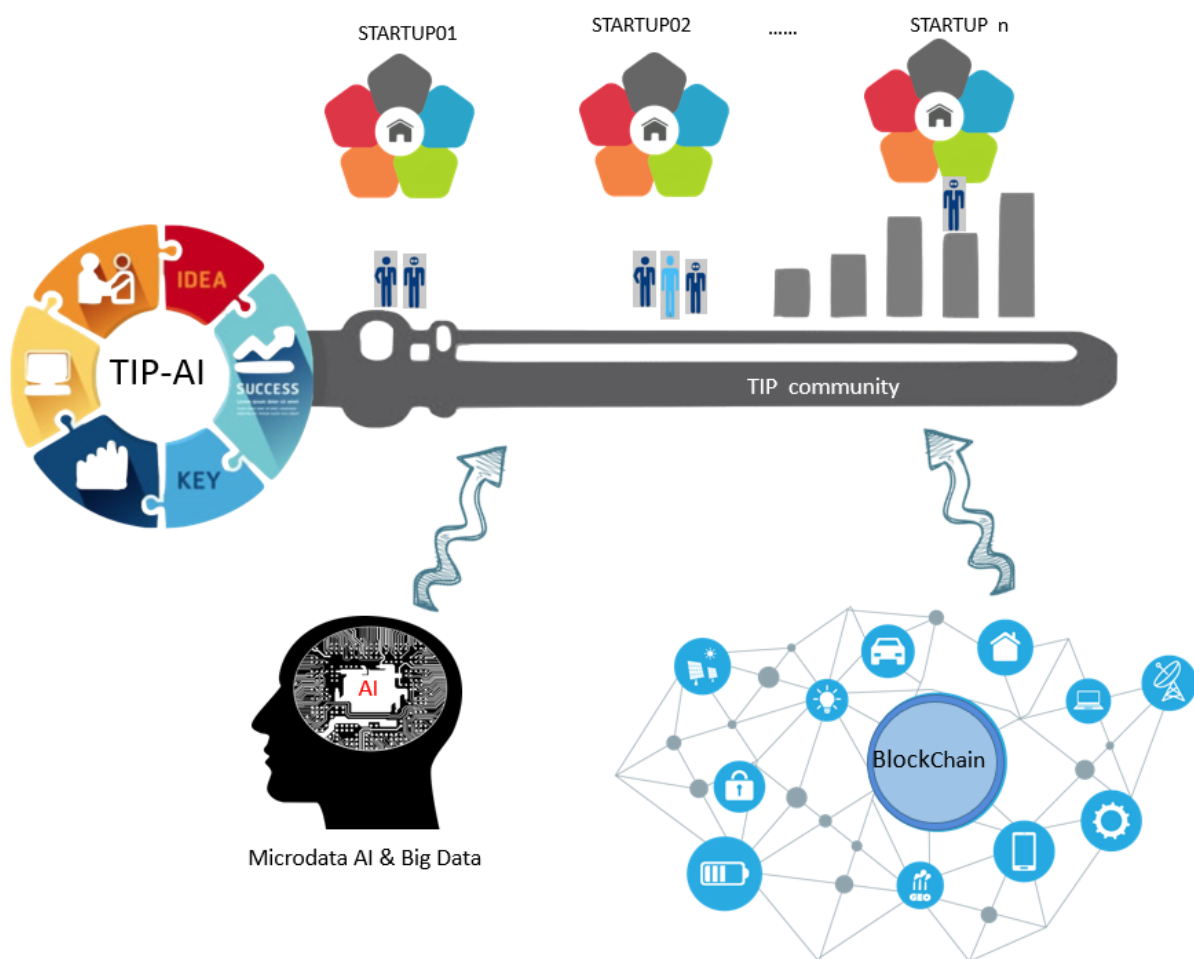
The TIP community has a very senior team of fund specialists. With years of experience in the blockchain technology and intellectual property industries, as well as professional standards of legal experts, the TIP community provides strong fund service consulting and asset security protection;

Incubator.tipchain.io, a professional-grade IP project incubator, and an industry-experienced incubator advisor throughout the escort.

This layer also includes the investment promotion market and exchanges outside the TIP community, which accelerates the conversion of value for the community's excellent IP projects.

4. Community business model of TIP

4.1 CBusiness architecture



The three-tier business architecture model consists of an upper-level IP ecosystem based on TIP Chain sub-chains for each StartUp company, a middle-level TIP community service platform that integrates micro-data artificial intelligence assistants of community members, and a TIP community including a high-performance blockchain system at the grass-roots level. The technical foundation layer of advanced AI service platform.

4.2 How do TIPC members get Top Intellectual Point

The ways that community members get Top Intellectual Point are :

- (1) There will be a certain scale of promotion activities in the early stage of the establishment of TIPC community. At this stage, there will be 1%~3% of the total amount of issue of Top Intellectual Point distributed to active members of community as promotion award. The bonus distributed at this stage will be appended or reduced depending on the promotion effect;
- (2) The Intellectual Property expert relies on his own excellent, high-quality IP rights to exchange for the Top Intellectual Point with corresponding level of IP. This channel highlights the TIPC community's recognition of the value of the contributors;
- (3) By investing in the community, the public investor earns Top Intellectual Point based on the proportion of the investment. This channel undertakes social investment;
- (4) By publishing their own unique idea on block chain innovation, ordinary members receive praise and Top Intellectual Point from other members of the community and vote;
- (5) The expert selects the excellent Intellectual Property to vote. Experts generally hold a higher proportion of votes, so the points obtained from this channel are also the most significant.
- (6) Members of the community can be exchanged for the Top Intellectual Point sold by other community members through the cryptocurrency trading platform. This is the most direct integration access channel and other asset conversion channels.
- (7) After the community members publish their ideas, they can transfer the ownership of the Intellectual Property to obtain a certain amount of the integration of the Intellectual Property in the community voting process. This channel increases the flexibility of Intellectual Property conversion;
- (8) The community organizer will organize the community construction and other incentive activities according to the community atmosphere, and some points will be distributed to the community members in return. This channel will reflect the cohesion of the community to a certain extent.

4.3 The selection process of high-quality intellectual property in the TIPC community

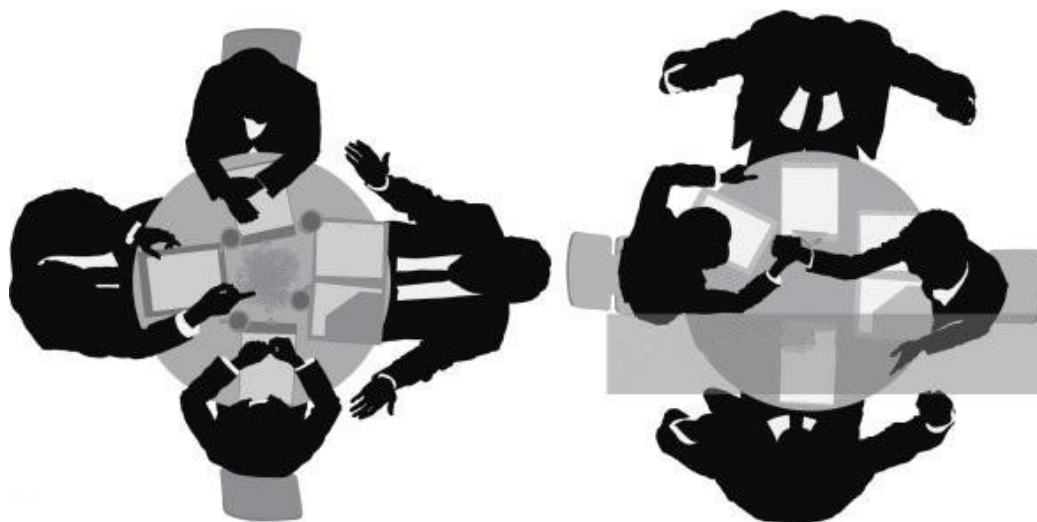
4.3.1 The vote

Any member of the TIPC community can offer their own ideas about Blockchain technology fairly and freely. We welcome and encourage any member to express their own ideas in a brainstorming manner, and the attitude of TIPC community in the Blockchain technology is always full of the passion and enthusiasm.

The vote in the TIPC community is the most free way of mass voting. There is only one principle: if you think it's good, vote for him!



The voting scenes of TIPC community can be changed at any time according to the demand of the market and business. It is diverse, flexible, and responsive to the needs of community members. The following is the form of voting in the early stages of the community:





The preliminary selection process is summarized as follows:

According to the function of innovation Idea, the expert group divide the Idea innovations that submitted by community members into groups. Taking the Idea copywriting from the top 5% of each category and write it as a professional IP document in an expert selection cycle, meanwhile, a brief public display is show in the whole community. The published content is the summary or some technical details of the copywriting provided by writer. During this period, the entire community members of TIPC can have in-depth discussions with the group including expert members and Idea copywriter, on the idea that they like. This series of publicity and discussion laid the foundation for the next TIP vote of shared permissions of IP, and every outstanding IP copy is accepted vote by anyone. Details of the vote are drawn up as follows:

- 1) All the voting process was completed through a smart contract, which ensured the transparency of the voting details;
- 2) Every outstanding IP only select the top four voters and share the ownership with IP provider, and the rest of the voters' Top Intellectual Point will return to the original output address through smart contracts.
- 3) The IP provider directly obtains the Top Intellectual Point of the four voters by a certain proportion;
- 4) The TIPC community shares the full ownership of IP with all outstanding IP ownership holders. For example, IP provider has 50% of IP ownership, while the TIPC community has 10% ownership, and the remaining 40% of IP ownership is allocated according to the percentage of votes cast by the four voters.

4.3.2 Transfer of IP ownership

The TIPC community has opened up the IP ownership transfer section, which includes Top Intellectual IP capacity transfer section and community member property transfer section.

Top Intellectual IP Capacity Transfer Section

This section is a investment channel that conversion IP into product. The TIPC community has made all the abstracts and voting of all outstanding IP available, which provides investors with comprehensive IP information. Investors can submit their investment intention to the TIPC

community according to their favorite IP project. If there are plenty investors, the TIPC community will make an IP transfer resolution after consulting with the IP owners.

The transfer process of IP ownership is also carried out through smart contract.

Community Member Property Transfer Section

Under the premise of compliance and legitimacy, the TIPC community allows the Top Intellectual Point transferring between members of the community, with any proportion of IP ownership. This section makes community members invest more flexible. Members can display their IP ownership in this section in the form of a minimum unit of 1%. It is then followed by a conversion between IP ownership and TIP by subsequent community members, based on transactions of smart contracts.

The detailed data on the vote of TIPC and the share transfer of equity income that mentioned above if for reference only, the actual operation will be more precise and coordinate. All vote, transfer process that involves TIPC are base on the smart contract of TIPC, and all steps of IP rights and interest are transparent. It accept the oversight of all members of the community.

There are various voting scenes in the TIPC community, which fully encourage community members to contribute ideas and to create a perfect and perfect and impeccable community. The final decision of high quality Intellectual Property in any scene is confirmed by members of the TIPC community expert group, in case of Intellectual Property flooding that bring negative impact on the whole community. The community will confiscate its entire Top Intellectual Points of community members of who disrupt the community rules seriously and bring an irreversible impact. The community builder setting up a supervise team which constituted by community members, and the confiscated Top Intellectual Point are allocated to members who contribute to community ecological construction.

4.4 Other application scenarios of Top Intellectual Point.

4.4.1 IP on the chain

TIPC has high performance block chain technology and as a basis, all members of the community, whether it is to publish new IP or to participate in the integration vote, are carried out on the chain. As the key protection object of property rights, excellent IP can obtain more professional rights protection on "chain". The newly proposed IP of community members must be "registered on the chain" to be recognized and accepted by the community platform.

The TIPC high-performance chain will do "on chain" dispose base on the required specifications of the block chain after the members submit IP copy, and charge a small number of Top Intellectual Point (such as 100TIP/IP copy).

In this way, it is not only an positive right protection for excellent IP, but also to increase the chain cost of inferior IP, which reduce the interference term and increase the harmony of the community.

4.4.2 Storage Permissions of TIPC

Excellent IP copywriting must be supported by a lot of images or video data to support its readability and IP authority. At this time, it is necessary to place a large number of data in the high-performance distributed database cluster of the TIPC. Quick and easy access to be obtained the copy, as a high performance database cluster for each address hash map with TIPC, it is not only to multiple backup data to provide strong security performance, and the data provide through full-text retrieval is more accurate and fast Because of the high quality and efficiency, it is more important to eliminate the interference of garbage data, TIP is also the threshold for data warehousing., such as the use of 1 TIP=100M/ month to obtain the space using right.

4.4.3 The Permission of Tracing and Viewing

The traceability of the Blockchain is also reflected in the IP permission data when the excellent IP on the chain is provided with the protection of ownership.

For example, one block chain technology expert or risk project investment expert think that one project has a promising future, because the TIPC is real-name chain, so these high value and traceable information can be clearly retrieved. This has undoubtedly provided an investment barometer for members of the community. In order to ensure that every data has been totally ran out of, without wasting the system resources, every data trace and view will receive a symbolic TIP (such as 0.001TIP/ time). This could reduces the risk of data was stolen massively. Each viewing can be used for a particular IP as a viewing condition, members can viewing all process start with the "on chain" to the viewing timing, including IP ownership distribution, IP rights transfer fee, holder information and other valuable information.

4.4.4 Incubator

TIPC community provides comprehensive project incubation platform: incubator.tipchain.io. Experts in the industry can provide all-round professional knowledge support for the project's incubators. Community members can use TIP to hire senior mentors for their projects to make the project more successful. Project incubation platform supports the establishment of proprietary forum, in order to attract the community members, increase fan loyalty of the project and forum participation. Meanwhile, the project side can maintain the activeness of project proprietary forum by rewarding TIP.

4. Management Standards of TIPC

Management standards of TIPC ensures the security, reliability and effectiveness of the TIPC Blockchain system, which mainly by quality, cost, time, resource utilization, system efficiency, confidentiality, integrity, availability, etc.

The object management standards of TIPC is the resources associated with the block chain, which mainly including participants, consensus mechanisms, nodes, application systems, algorithms, infrastructure, and data. To plan and deal with the information and related resources of Blockchain; Determine the processing of Blockchain by planning and organization, collection and implementation, delivery and support, security and monitoring, etc., and give each process detailed control objectives, audit guidelines and evaluation methods.

TIPC's Blockchain technology needs to set the core management structure according to the business and regulatory requirements of the industry to ensure compliance with regulatory requirements, meet international standards and meet the basic rules of all industries.

TIPC will adhere to the following five management principles:

- 1) Legality and compliance: comply with relevant national laws and regulations and regulatory requirements, providing technical support for regulatory audit requirements;
- 2) Traceability: both business and activities are documented, traceable and auditable;
- 3) Security: take all necessary security measures to ensure the security of information on the chain, such as assets and transactions, and prevent attacks;
- 4) Privacy protection: protect the privacy of users on the chain and prevent the leakage of users' privacy;
- 5) Business-oriented: the applicable business scenarios need to be considered primarily when technology, design and development are driven by demand. It is usually needs to rearrange the process of the original business. On the one hand, the characteristics of the Blockchain

technology can be fully utilized while maintaining compliance, on the other hand, it should also consider how to bring improvement and value creation for the business.

Conclusion

TIPC community team realize that Blockchain is not just a technology, but The socialization of consensus on trust. We follow the principle of mutual trust in the Blockchain that all data can be traced back, combining artificial intelligence technology, and actively develop the preciseness that everything is unchangeable.

Top Intellectual Point (TIP) Allocation Scheme

Total amount of TIP: 1 billion

Global private placements: 10%

Promotion fee: 20%

TIP transfer and purchase: 40%

The operation of TIP community: 20%

Team: 10%

Initial private placement: 2% (Total: 20 million)

TIP's opening price on overseas exchanges: About RMB 1 per point

Explanation of term

1. Distributed

A distributed system is a system consisting of a set of computer nodes that communicate through the network and work together to accomplish common tasks.

2. Consensus Mechanism

The consensus mechanism is a process in a distributed system that is used to achieve data consistency across all nodes and agree on a proposal in a network that designs multiple unreliable nodes.

3. PoW

The Proof Of Work is mentioned in bitcoins firstly. The digital currency miners obtain the block award by calculating the current block chain by random Hash. The feature of PoW is that hashing is random, difficult to falsify, and easy to be verified. But on the other hand, the miners wasted a lot of energy because of the competition of Hash calculation.

4. PoS

The Proof of Stake consensus algorithm is the alternative of PoW. Determine the probability of obtaining block bookkeeping power according to the proportion of equity in the node. The more equity, the greater the opportunity to get the right of recording the block.

5. Smart Contract

Smart contract was first proposed by Nick Szabo in 1994. It is a computer contract designed to disseminating, validating or enforcing contracts by informatization. It allows credible transactions without a third party, which can be tracked and irreversible. The purpose of smart contract is to provide better security than traditional contract methods and reduce other transaction costs associated with the contract.

6. Neural network:

A high-performance computing network system that simulates the operation of the human brain.

7. Bayes' theorem

Bayes's theorem is to make inferences and decisions about the uncertainty of random events. It is necessary to estimate the probabilities of various conclusions. In this case, the AI is added with an involuntary uncertainty change factor.

8. Evolutionism

The derivation theory of use and disuse theory in AI.

9. Analogy inference

Here refers to artificial intelligence cross-domain autonomous learning.

10. Random forest

Refers to an infinite number of floating slots generated in analogy.

11. Sonar regression

Active CI to create information.

12. AI dual-core algorithm

Microdata+big data is a huge number of data calculations that surpass the value of existing AI data.

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